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/*  
 * C Program to Perform Merge Sort using Recursion and Functions  
 */
```

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
// merge function
```

```
void Merge(int arr[], int left, int mid, int right)
```

```
{
```

```
    int i, j, k;
```

```
    int size1 = mid - left + 1;
```

```
    int size2 = right - mid;
```

```
    // created temporary array
```

```
    int Left[size1], Right[size2];
```

```
    // copying the data from arr to temporary array
```

```
    for (i = 0; i < size1; i++)
```

```
        Left[i] = arr[left + i];
```

```
    for (j = 0; j < size2; j++)
```

```
        Right[j] = arr[mid + 1 + j];
```

```
    // merging of the array
```

```

i = 0; // initial index of first subarray
j = 0; // initial index of second subarray
k = left; // initial index of parent array
while (i < size1 && j < size2)
{
    if (Left[i] <= Right[j])
    {
        arr[k] = Left[i];

        i++;
    }
    else
    {
        arr[k] = Right[j];

        j++;
    }
    k++;
}

// copying the elements from Left[], if any
while (i < size1)
{
    arr[k] = Left[i];

    i++;

    k++;
}

```

```
// copying the elements from Right[], if any
while (j < size2)
{
    arr[k] = Right[j];

    j++;

    k++;
}
}
```

```
//merge sort function
void Merge_Sort(int arr[], int left, int right)
{
    if (left < right)
    {

        int mid = left + (right - left) / 2;

        // recursive calling of merge_sort
        Merge_Sort(arr, left, mid);
        Merge_Sort(arr, mid + 1, right);

        Merge(arr, left, mid, right);
    }
}
```

```
// driver code

int main()

{

    int size;

    printf("Enter the size: ");

    scanf("%d", &size);


    int arr[size];

    printf("Enter the elements of array: ");

    for (int i = 0; i < size; i++)

    {

        scanf("%d", &arr[i]);

    }


    Merge_Sort(arr, 0, size - 1);


    printf("The sorted array is: ");

    for (int i = 0; i < size; i++)

    {

        printf("%d ", arr[i]);

    }

    printf("\n");

    return 0;

}
```